

Chapter 8: Self Test

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Roadmap

- **Problems 1–3:** analytic tests, isolated singularities, residues, and the poles of $\Gamma(z)$.
- **Problems 4–5:** direct Gamma/Beta evaluations and contour-integral normalization constants.
- **Problem 6:** precise definitions, including branch cuts, analytic continuation, Fuchs' theorem, and Bessel behavior near the origin.
- **Problem 7:** Laurent expansion and residue bookkeeping for an essential singularity multiplied by simple poles.
- **Problems 8–10:** Hermite and Chebyshev equations, series termination, recursion, and orthogonality.

Where we are going

This chapter is the book's self-test. The slides keep every problem from the text and give corrected, worked solutions.

- Problems 1–3 check analytic functions, singularities, residues, and Gamma poles.
- Problems 4–5 evaluate real integrals with Gamma/Beta and contour methods.
- Problem 6 asks for precise short definitions.
- Problem 7 combines an essential singularity with simple poles.
- Problems 8–10 revisit Hermite and Chebyshev polynomials by Frobenius and orthogonality.

Problems 1–3

1. Answer the following.

- (a) True or false: a function is analytic at $z = z_0$ only if it has a Taylor series representation at z_0 .
- (b) True or false: if a function is analytic at $z = z_0$, then its derivative must be analytic at $z = z_0$.
- (c) True or false: all analytic functions satisfy the Cauchy–Riemann conditions.
- (d) True or false: the function $\cos z$ has an essential singularity at infinity.
- (e) Each function has a singularity at the origin. Classify it: $\sin(1/z)$, $z/\sin z$, and $1/\sin z$.
- (f) Compute $\text{Res}[e^z/z^5, 0]$.

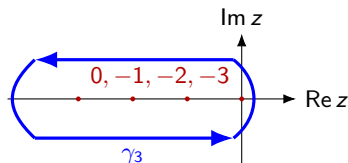
2.

- (a) Locate all poles of $1/\sin z$.
- (b) Compute $\oint_{|z|=10} \frac{1}{\sin z} dz$, with the contour counterclockwise.

Problems 1–3 (cont.)

3.

- (a) Locate all poles of the Gamma function $\Gamma(z)$.
- (b) Compute the residues at those poles.
- (c) Let γ_N be a counterclockwise contour enclosing $0, -1, \dots, -N$ and no other poles of Γ . Compute $\oint_{\gamma_N} \Gamma(z) dz$ and then take the limit $N \rightarrow \infty$.



Problems 4–7

4. Compute

$$\int_0^{\infty} e^{-x} x^{2/3} dx.$$

5. Compute

$$\int_0^{\infty} \frac{\sqrt{x}}{x^2 + 1} dx.$$

6. Short answer questions.

- (a) Give an example of a function with a branch cut in the complex plane.
- (b) Define removable singularity, regular singularity, and essential singularity; give an example of each.
- (c) Explain analytic continuation.
- (d) State when Fuchs' theorem assures a Frobenius solution for $p_0(x)u'' + p_1(x)u' + p_2(x)u = 0$.
- (e) State when the Sturm–Liouville operator $Lu = \frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + q(x)u$ is self-adjoint.
- (f) Describe $J_m(x)$ for $x \ll 1$.

Problems 4–7 (cont.)

7. Consider

$$f(z) = \frac{e^{-1/z^2}}{1+z^2}.$$

- (a) Determine and classify all singularities.
- (b) Determine the residue at $z = 0$.
- (c) Determine $\int_0^\infty f(x) dx$.

Problems 8–10

8. Use the method of Frobenius to solve the Hermite equation

$$H''(x) - 2xH'(x) + 2nH(x) = 0.$$

9. Use the method of Frobenius to solve the Chebyshev equation

$$(1 - x^2)T_n'' - xT_n' + n^2T_n = 0.$$

10. Let $H_n(x)$ be a Hermite polynomial and let m, n be nonnegative integers. Compute

$$\int_{-\infty}^{\infty} xe^{-x^2} H_n(x) dx \quad \text{and} \quad \int_{-\infty}^{\infty} xe^{-x^2} H_n(x) H_m(x) dx.$$

Solution 1: analytic functions and singularities

- (a) **True.** A complex analytic function has a convergent Taylor series in a neighborhood of the point.
- (b) **True.** Analytic functions are complex differentiable to all orders, so the derivative is analytic on the same neighborhood.
- (c) **True.** If $f = u + iv$ is analytic, then $u_x = v_y$ and $u_y = -v_x$.
- (d) **True.** A function has a singularity at infinity when $f(1/w)$ has a singularity at $w = 0$. Since $\cos(1/w)$ has infinitely many negative powers in its Laurent series, infinity is an essential singularity of $\cos z$.
- (e) The classifications at 0 are:

$$\sin(1/z) \text{ essential,} \quad \frac{z}{\sin z} \text{ removable,} \quad \frac{1}{\sin z} \text{ simple pole.}$$

The removable case has limiting value 1 because $\sin z \sim z$.

Solution 1: analytic functions and singularities (cont.)

(f) Since $e^z = \sum_{k=0}^{\infty} z^k/k!$,

$$\frac{e^z}{z^5} = \sum_{k=0}^{\infty} \frac{z^{k-5}}{k!}.$$

The residue is the coefficient of z^{-1} , hence $k = 4$ and

$$\text{Res}\left[\frac{e^z}{z^5}, 0\right] = \frac{1}{4!} = \frac{1}{24}.$$

Solution 2: poles and residues of $\csc z$

The poles of $1/\sin z$ occur at the zeros of $\sin z$, namely

$$z = n\pi, \quad n \in \mathbb{Z}.$$

They are simple poles because $\sin'(n\pi) = \cos(n\pi) = (-1)^n \neq 0$. For a simple zero of g , the residue of $1/g$ is $1/g'(z_0)$, so

$$\operatorname{Res}\left[\frac{1}{\sin z}, n\pi\right] = \frac{1}{\cos(n\pi)} = (-1)^n.$$

The circle $|z| = 10$ contains exactly $n = -3, -2, -1, 0, 1, 2, 3$, because $3\pi < 10 < 4\pi$. The residue sum is

$$\sum_{n=-3}^3 (-1)^n = -1 + 1 - 1 + 1 - 1 + 1 - 1 = -1.$$

For counterclockwise orientation,

$$\oint_{|z|=10} \frac{1}{\sin z} dz = 2\pi i(-1) = -2\pi i. \quad (8.2.1)$$

If the same circle were traversed clockwise, the answer would change sign.

Solution 3: poles and residues of Γ

The Gamma function has simple poles at

$$z = 0, -1, -2, -3, \dots$$

To compute the residues, use Euler's reflection formula in the form

$$\Gamma(z) = \frac{\pi}{\Gamma(1-z)\sin(\pi z)}.$$

Near $z = -n$, the factor $\sin(\pi z)$ has a simple zero. Let $F(z) = \pi/\Gamma(1-z)$ and $G(z) = \sin(\pi z)$. Then

$$F(-n) = \frac{\pi}{\Gamma(n+1)} = \frac{\pi}{n!}, \quad G'(-n) = \pi \cos(-n\pi) = \pi(-1)^n.$$

Therefore

$$\operatorname{Res}[\Gamma(z), -n] = \frac{F(-n)}{G'(-n)} = \frac{\pi/n!}{\pi(-1)^n} = \frac{(-1)^n}{n!}. \quad (8.2.2)$$

The factor of π cancels.

Solution 3: poles and residues of Γ (cont.)

Let γ_N enclose the finite set of poles $0, -1, \dots, -N$ counterclockwise. Then the residue theorem gives the finite identity

$$\oint_{\gamma_N} \Gamma(z) dz = 2\pi i \sum_{n=0}^N \frac{(-1)^n}{n!}.$$

Therefore, for a contour family whose enclosed finite pole set grows to the left,

$$\lim_{N \rightarrow \infty} \oint_{\gamma_N} \Gamma(z) dz = 2\pi i \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = 2\pi i e^{-1}.$$

Solutions 4–5: two real integrals

For Problem 4, compare directly with the Gamma integral:

$$\Gamma(a) = \int_0^{\infty} e^{-x} x^{a-1} dx.$$

Here $a - 1 = 2/3$, so $a = 5/3$ and

$$\int_0^{\infty} e^{-x} x^{2/3} dx = \Gamma\left(\frac{5}{3}\right). \quad (8.2.3)$$

For Problem 5, use the standard beta-contour value

$$\int_0^{\infty} \frac{x^{a-1}}{1+x^b} dx = \frac{\pi}{b} \csc\left(\frac{\pi a}{b}\right), \quad 0 < a < b.$$

With $x^{1/2} = x^{3/2-1}$, we have $a = 3/2$ and $b = 2$. Hence

$$\int_0^{\infty} \frac{\sqrt{x}}{x^2+1} dx = \frac{\pi}{2} \csc\left(\frac{3\pi}{4}\right) = \frac{\pi}{2} \sqrt{2} = \frac{\pi}{\sqrt{2}}. \quad (8.2.4)$$

Solutions 4–5: two real integrals (cont.)

The same answer follows from the book's keyhole contour for

$$f(z) = \frac{z^{1/2}}{1+z^2}$$

with the positive real axis as the branch cut and $0 < \arg z < 2\pi$. The poles inside are $z = i$ and $z = -i$, with branch values $i^{1/2} = e^{i\pi/4}$ and $(-i)^{1/2} = e^{3\pi i/4}$. Hence

$$\oint_C f(z) dz = 2\pi i \left(\frac{e^{i\pi/4}}{2i} + \frac{e^{3\pi i/4}}{-2i} \right) = \pi\sqrt{2}.$$

The *ML*-estimate gives

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{R^{1/2}}{R^2 - 1} 2\pi R = O(R^{-1/2}) \rightarrow 0,$$

and

$$\left| \int_{C_\varepsilon} f(z) dz \right| \leq \frac{\varepsilon^{1/2}}{1 - \varepsilon^2} 2\pi\varepsilon = O(\varepsilon^{3/2}) \rightarrow 0.$$

Solutions 4–5: two real integrals (cont.)

On the two banks of the cut,

$$z = r \quad \text{on the upper bank}, \quad z = re^{2\pi i} \quad \text{on the lower bank},$$

so $z^{1/2}$ changes by $e^{\pi i} = -1$, and the lower bank is traversed in the opposite direction. The two bank integrals therefore add:

$$\int_{\gamma_1} f \, dz + \int_{\gamma_2} f \, dz = 2 \int_0^\infty \frac{\sqrt{r}}{1+r^2} \, dr.$$

Thus $2I = \pi\sqrt{2}$, so $I = \pi/\sqrt{2}$.

Solution 6: short answers

- (a) A standard example is $\log z$, made single-valued by cutting the plane along a ray such as the negative real axis and fixing $\arg z \in (-\pi, \pi)$.
- (b) A *removable singularity* is a point where redefining the value makes the function analytic; example $z/\sin z$ at 0. A *regular singularity* here means a pole of finite order; example $1/z^2$ at 0. An *essential singularity* has infinitely many negative Laurent powers; example $e^{1/z}$ at 0.
- (c) Analytic continuation extends a function from one domain to a larger connected domain by matching it with analytic functions on overlapping open sets. The identity theorem makes the extension unique when it exists.
- (d) Put the ODE in standard form $u'' + P(x)u' + Q(x)u = 0$. Fuchs' theorem gives a Frobenius solution at x_0 when $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0 .
- (e) The operator $Lu = (pu')' + qu$ is formally self-adjoint. It is genuinely self-adjoint on a chosen domain when the boundary term $p(vu' - v'u)|_a^b$ vanishes for all allowed u, v ; Dirichlet, Neumann, and periodic boundary conditions are common examples.
- (f) For $x \ll 1$,

$$J_m(x) \sim \frac{1}{m!} \left(\frac{x}{2}\right)^m \quad (m = 0, 1, 2, \dots).$$

Solution 7: $e^{-1/z^2}/(1+z^2)$

The function

$$f(z) = \frac{e^{-1/z^2}}{1+z^2}$$

has an essential singularity at $z = 0$, because e^{-1/z^2} has infinitely many negative even powers. It also has simple poles at $z = i$ and $z = -i$, since $1 + z^2 = (z - i)(z + i)$ and e^{-1/z^2} is analytic and nonzero there.

For the residue at 0, expand in the punctured unit disk:

$$e^{-1/z^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} z^{-2k}, \quad \frac{1}{1+z^2} = \sum_{j=0}^{\infty} (-1)^j z^{2j}.$$

Every power in the product is even, z^{2j-2k} , so no z^{-1} term appears:

$$\operatorname{Res}[f, 0] = 0. \tag{8.2.5}$$

Solution 7: $e^{-1/z^2}/(1+z^2)$ (cont.)

For the real integral, a tempting upper-semicircle shortcut is not justified as stated: although the real-axis value at 0 is removable, e^{-1/z^2} has an essential singularity at $z = 0$ in the complex plane. A small upper indentation $z = \varepsilon e^{i\theta}$ has

$$\left| e^{-1/z^2} \right| = \exp\left(-\frac{\cos 2\theta}{\varepsilon^2}\right),$$

which blows up near $\theta = \pi/2$. Hence the small-arc contribution cannot be discarded, and the contour argument that keeps only the pole at i is not a proof.

Solution 7: $e^{-1/z^2}/(1+z^2)$ (cont.)

Use the corrected real-variable reduction. Set $u = 1/x$. Then $dx = -du/u^2$ and

$$1 + x^2 = 1 + \frac{1}{u^2} = \frac{1 + u^2}{u^2}.$$

Thus

$$\int_0^\infty \frac{e^{-1/x^2}}{1+x^2} dx = \int_\infty^0 e^{-u^2} \frac{u^2}{1+u^2} \frac{-du}{u^2} = \int_0^\infty \frac{e^{-u^2}}{1+u^2} du.$$

Using the complementary error function

$$\operatorname{erfc}(a) = \frac{2}{\sqrt{\pi}} \int_a^\infty e^{-t^2} dt,$$

derive the needed parameter integral as follows. For $a > 0$,

$$\frac{1}{u^2 + a^2} = \int_0^\infty e^{-(u^2 + a^2)s} ds,$$

Solution 7: $e^{-1/z^2}/(1+z^2)$ (cont.)

so

$$\begin{aligned} I(a) &= \int_0^\infty \frac{e^{-u^2}}{u^2 + a^2} du \\ &= \frac{\sqrt{\pi}}{2} \int_0^\infty e^{-a^2 s} (1+s)^{-1/2} ds \\ &= \frac{\sqrt{\pi} e^{a^2}}{a} \int_a^\infty e^{-t^2} dt = \frac{\pi}{2a} e^{a^2} \operatorname{erfc}(a). \end{aligned}$$

Taking $a = 1$ gives

$$\int_0^\infty f(x) dx = \frac{\pi e}{2} \operatorname{erfc}(1). \quad (8.2.6)$$

Solution 8: Hermite by Frobenius

The point $x = 0$ is ordinary, but following the book's Frobenius setup take

$$H(x) = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

Substitute in $H'' - 2xH' + 2nH = 0$, shift the H'' sum by $k \mapsto k + 2$, and use $a_{-1} = a_{-2} = 0$. The coefficient relation is

$$(k + r + 2)(k + r + 1)a_{k+2} + 2(n - k - r)a_k = 0.$$

The $k = -2$ term gives the indicial equation $r(r - 1) = 0$, so $r = 0$ or $r = 1$. For the even chain $r = 0$,

$$a_{k+2} = \frac{2(k - n)}{(k + 2)(k + 1)} a_k. \quad (8.2.7)$$

The even and odd chains are independent. When n is a nonnegative integer, the chain of the same parity as n terminates, producing a polynomial.

Solution 8: Hermite by Frobenius (cont.)

Starting with a_0 , the first even coefficients are

$$a_2 = -na_0, \quad a_4 = \frac{2^2 n(n-2)}{4!} a_0, \quad a_6 = -\frac{2^3 n(n-2)(n-4)}{6!} a_0.$$

In general,

$$a_{2k} = (-1)^k \frac{2^k}{(2k)!} \prod_{j=0}^{k-1} (n-2j) a_0,$$

with the empty product equal to 1. For even n , the product terminates when a factor becomes zero; reverse the finite sum to obtain the standard normalization below. The odd chain follows similarly from a_1 .

Solution 8: Hermite by Frobenius (cont.)

With the standard leading coefficient 2^n , the polynomial is

$$H_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j (2x)^{n-2j} \frac{n!}{j!(n-2j)!}. \quad (8.2.8)$$

This is the Hermite polynomial used in the oscillator problem.

Solution 9: Chebyshev by Frobenius

Again $x = 0$ is ordinary. Put $T = \sum_{k=0}^{\infty} a_k x^k$ in

$$(1 - x^2)T'' - xT' + n^2 T = 0.$$

The coefficient of x^k is

$$(k+2)(k+1)a_{k+2} - k(k-1)a_k - ka_k + n^2 a_k = 0,$$

or

$$a_{k+2} = \frac{k^2 - n^2}{(k+2)(k+1)} a_k = -\frac{(n-k)(n+k)}{(k+2)(k+1)} a_k. \quad (8.2.9)$$

The recurrence preserves parity. If n is even, the even chain terminates at $k = n$; if n is odd, the odd chain terminates at $k = n$.

Solution 9: Chebyshev by Frobenius (cont.)

Equivalently, in the book's Frobenius notation $T_n(x) = \sum_{m=0}^{\infty} a_m x^{m+k}$. The lowest power gives the indicial equation

$$k(k-1)a_0 = 0,$$

so $k = 0$ or $k = 1$. For $k = 0$, after shifting the differentiated sum by two powers,

$$m(m-1)a_m + [n^2 - (m-2)^2]a_{m-2} = 0,$$

or, with $m \mapsto m+2$,

$$a_{m+2} = -\frac{(n-m)(n+m)}{(m+2)(m+1)}a_m.$$

Thus

$$a_2 = -\frac{n^2}{2!}a_0, \quad a_4 = \frac{n^2(n^2-2^2)}{4!}a_0, \quad a_6 = -\frac{n^2(n^2-2^2)(n^2-4^2)}{6!}a_0,$$

and the analogous odd chain starts from a_1 .

Solution 9: Chebyshev by Frobenius (cont.)

Choosing the normalization $T_n(1) = 1$ gives the Chebyshev polynomial

$$T_n(x) = \cos(n \arccos x). \quad (8.2.10)$$

For $n \geq 1$, the explicit series is

$$T_n(x) = \frac{n}{2} \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (n-j-1)!}{j!(n-2j)!} (2x)^{n-2j}. \quad (8.2.11)$$

The first cases are $T_0 = 1$, $T_1 = x$, $T_2 = 2x^2 - 1$, and $T_3 = 4x^3 - 3x$.

Solution 10: Hermite weighted integrals

Use the recurrence

$$H_{n+1} = 2xH_n - 2nH_{n-1},$$

which rearranges to

$$xH_n = \frac{1}{2}H_{n+1} + nH_{n-1}. \quad (8.2.12)$$

Also use the Hermite norm

$$\int_{-\infty}^{\infty} H_k(x)^2 e^{-x^2} dx = 2^k k! \sqrt{\pi}.$$

For Problem 10(a), take $H_0 = 1$ in the second factor:

$$\int_{-\infty}^{\infty} x e^{-x^2} H_n(x) dx = \frac{1}{2}(H_{n+1}, H_0)_w + n(H_{n-1}, H_0)_w.$$

Only $n = 1$ survives, and the value is $\sqrt{\pi}$:

$$\int_{-\infty}^{\infty} x e^{-x^2} H_n(x) dx = \sqrt{\pi} \delta_{n1}. \quad (8.2.13)$$

Solution 10: Hermite weighted integrals (cont.)

For Problem 10(b), multiply (8.2.12) by $H_m e^{-x^2}$ and integrate:

$$\int_{-\infty}^{\infty} x e^{-x^2} H_n H_m dx = \frac{1}{2} \int H_{n+1} H_m e^{-x^2} dx + n \int H_{n-1} H_m e^{-x^2} dx.$$

Orthogonality leaves only $m = n + 1$ and $m = n - 1$:

$$\int_{-\infty}^{\infty} x e^{-x^2} H_n H_m dx = \begin{cases} 2^n (n+1)! \sqrt{\pi}, & m = n + 1, \\ 2^{n-1} n! \sqrt{\pi}, & m = n - 1, \\ 0, & \text{otherwise.} \end{cases} \quad (8.2.14)$$

The plus sign in (8.2.12) is essential: it comes directly from the standard recurrence.

Summary

- Residues are coefficients of $(z - z_0)^{-1}$; for a simple zero $g(z_0) = 0$, $\text{Res}[f/g, z_0] = f(z_0)/g'(z_0)$.
- Corrected self-test constants:

$$\oint_{|z|=10} \csc z \, dz = -2\pi i, \quad \text{Res}[\Gamma, -n] = \frac{(-1)^n}{n!}, \quad \lim_{N \rightarrow \infty} \oint_{\gamma_N} \Gamma(z) \, dz = 2\pi i e^{-1}.$$

- The keyhole integral gives $\int_0^\infty \sqrt{x}/(1+x^2) \, dx = \pi/\sqrt{2}$.
- Hermite and Chebyshev both come from ordinary-point series recurrences; polynomial termination is the quantization condition.